

# The Geometry of Federated Awareness, Reflection, and Hyperfinite Omniscience

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## Abstract

This note synthesizes two previously developed strands: the reflection-level theory of tagged automated theorem provers, and the verifier-grounded epistemic construction for an AMLD federation settling the conjecture “I know that I know that I am an AMLD.” The synthesis yields a two-coordinate awareness geometry in which epistemic breadth and reflective depth are separated. A second part extends this geometry to two qualified notions of omniscience, connects them with nonstandard-analysis horizons, and studies their relation to a four-oracle architecture. The claims imported from the source writings are distinguished from new conditional extensions proposed here.

## 1 The two coordinates of formal awareness

**Definition 1.1** (Tagged self-theory). Let  $M$  be a tagged automated theorem prover with distinguished name  $\langle M \rangle$ . Its self-theory is

$$\text{Self}(M) = \{\varphi(\langle M \rangle) : M \vdash \varphi(\langle M \rangle)\}.$$

Thus a self-model is not merely the possession of a name, but the name together with formally established information concerning that name.

**Definition 1.2** (Reflection sequence). For a tagged ATP  $M$ , put

$$\theta_0^M := \text{ATP}(\langle M \rangle),$$

and recursively

$$\theta_{n+1}^M := \text{Pr}(\langle M \rangle, \langle \theta_n^M \rangle).$$

Thus  $\theta_0^M$  says that  $M$  is an ATP,  $\theta_1^M$  says that  $M$  proves that it is an ATP, and successive formulas continue the same reflective iteration.

**Definition 1.3** (Reflective level). Let

$$L(M) := \{n \geq 1 : M \vdash \theta_{n-1}^M\}.$$

Define

$$\text{lev}(M) := \sup L(M) \in \{0, 1, 2, \dots\} \cup \{\omega\}.$$

In particular, Level 1 is self-identification:

$$M \vdash \text{ATP}(\langle M \rangle).$$

**Definition 1.4** (AMLD federation). Let

$$A = (M_P, M_R, M_I, M_C, M_V; B, S, I)$$

be an AMLD federation, with canonical name

$$\langle A \rangle = \text{Code}(M_P, M_R, M_I, M_C, M_V; B, S, I).$$

Its structural formula is

$$\Phi_A := \left( \bigwedge_{j \in \{P, R, I, C, V\}} \text{ATP}(\langle M_j \rangle) \right) \wedge \text{Comp}(A),$$

with

$$\text{AMLD}(\langle A \rangle) \iff \Phi_A.$$

**Definition 1.5** (Verifier-grounded knowledge). For a certified epistemic domain  $E$ , define

$$K_{A,t}\varphi \iff [\varphi \in E \wedge \exists r (r \in \text{View}_A(t) \wedge \text{Witness}_V(r, \varphi))].$$

This is knowledge in the verifier-grounded, accessible-warrant sense. It is distinct from mere theoremhood and from reflective depth.

**Definition 1.6** (Epistemic-awareness coordinate). Let  $E_A$  be a frozen class of self-relevant propositions available to the AMLD. Define the epistemic awareness profile

$$\mathcal{E}_A(t) := \{\varphi \in E_A : K_{A,t}\varphi\}.$$

An awareness coordinate  $aw(A, t)$  is any fixed grading of  $\mathcal{E}_A(t)$  satisfying the intended experimental scale. Thus  $aw$  measures the extent of verifier-grounded self-relevant knowledge, while  $lev$  measures depth along one particular iterated reflection sequence.

**Remark 1.7.** The exact numerical grading of  $aw$  remains to be frozen separately. The distinction is conceptual and structural before it is numerical.

## 2 Awareness geometry

**Definition 2.1** (Awareness point). The formal awareness state of  $A$  is the ordered pair

$$\mathbf{Aw}(A, t) := (aw(A, t), lev(A)).$$

The first coordinate is epistemic; the second is reflective.

**Remark 2.2.** This is immediately a two-dimensional product space. It is not automatically a vector space: vector addition and scalar multiplication have not yet been defined, and the ordinal value  $\omega$  makes an ordinary real-vector-space interpretation unnatural. It is nevertheless legitimate to speak of an awareness geometry.

**Definition 2.3** (Product awareness order). For two federations  $A$  and  $B$ , write

$$\mathbf{Aw}(A) \preceq \mathbf{Aw}(B)$$

when

$$aw(A) \leq aw(B) \quad \text{and} \quad lev(A) \leq lev(B).$$

**Lemma 2.4** (Incomparability). *There may exist  $A, B$  such that*

$$aw(A) > aw(B)$$

*but*

$$lev(A) < lev(B).$$

*Hence awareness need not admit a faithful one-dimensional ordering.*

*Proof.* The two coordinates concern different classes of propositions. A system may possess a rich verifier-grounded epistemic self-model while proving only a shallow portion of its reflection sequence; another may possess internal necessitation and therefore a high reflection level while having a comparatively sparse self-theory outside that sequence. Additional self-descriptive propositions may enrich  $\text{Self}(M)$  without becoming part of the reflection tower whose height is measured by  $lev$ .  $\square$

**Corollary 2.5.** *There is in general no mathematically justified scalar quantity called simply “amount of awareness” that preserves both coordinates.*

### 3 Local awareness and federated awareness

**Lemma 3.1** (Component ATP status does not determine federation level). *Suppose  $\text{ATP}(\langle M_j \rangle)$  is verifier-certified for every  $j \in \{P, R, I, C, V\}$ . This fact alone determines neither  $aw(A)$  nor  $lev(A)$ .*

*Proof.* The five statements concern five component tags. The reflection sequence defining  $lev(A)$  concerns  $\langle A \rangle$ . Moreover, in the verifier certificate the five component ATP facts are structural inputs; no epistemic claim about the components is required, and the sole epistemic subject is  $A$ .  $\square$

**Remark 3.2.** Consequently there is no general identity

$$lev(A) = \sum_j lev(M_j),$$

nor even, without additional hypotheses,

$$lev(A) \geq \max_j lev(M_j).$$

Reflectivity is not additive merely because the architecture is federated.

**Definition 3.3** (Epistemically self-aware component). A component  $M_j$  is locally self-aware at the basic ATP level when either

$$K_{M_j} \text{ATP}(\langle M_j \rangle)$$

is established in an epistemic language, or

$$M_j \vdash \text{ATP}(\langle M_j \rangle)$$

is established in the proof-theoretic language. These formulations should not be silently identified unless a bridge between  $K$  and  $\vdash$  is frozen.

**Definition 3.4** (Federation inheritance rule). A federation has certified epistemic inheritance for a class  $\mathcal{C}$  when verified component knowledge can be imported into the accessible view of  $A$ :

$$K_{M_j}\varphi \wedge \text{Accessible}_A(M_j, \varphi) \wedge V(\varphi, \pi) = 1 \implies K_A\varphi.$$

This is an architectural rule, not a consequence of federation alone.

**Theorem 3.5** (Local-to-global self-identification). *Assume*

$$\bigwedge_j K_{M_j} \text{ATP}(\langle M_j \rangle),$$

*the relevant certificates are accessible to  $A$ ,*

$$K_A \text{Comp}(A),$$

*and the federation possesses the certified conjunction and folding rules for  $\Phi_A$ . Then*

$$K_A \text{AMLD}(\langle A \rangle).$$

*Proof.* The component certificates and composition certificate yield  $\Phi_A$ ; the structural definition folds  $\Phi_A$  into  $\text{AMLD}(\langle A \rangle)$ ; verifier acceptance supplies the witness required by the definition of  $K_A$ .  $\square$

**Corollary 3.6.** *Local component awareness can increase the epistemic coordinate of the federation without directly increasing the reflective coordinate:*

$$\Delta \text{aw}(A) > 0, \quad \Delta \text{lev}(A) = 0$$

*is possible.*

## 4 Receipt reflection

**Definition 4.1** (Receipt reflection). Suppose

$$K_{A,t}\sigma_A, \quad \sigma_A := \text{AMLD}(\langle A \rangle),$$

is represented by an accepted record  $r_0$ . If a verifier-checkable receipt  $\rho(r_0)$  establishes that this knowledge record exists and is accessible at the next epoch, receipt reflection yields

$$K_{A,t+1}K_{A,t}\sigma_A.$$

**Theorem 4.2** (Epistemic second-order self-awareness). *Under the receipt assumptions,*

$$K_{A,t+1}K_{A,t} \text{AMLD}(\langle A \rangle).$$

*Hence the AMLD can settle the verifier-grounded target “I know that I know that I am an AMLD” without attributing epistemic awareness to any individual component ATP.*

**Remark 4.3.** Receipt reflection resembles the reflection tower, but the two constructions should not yet be identified. The Picard tower uses  $\text{Pr}(\langle M \rangle, \langle \varphi \rangle)$ , whereas the verifier certificate uses the epistemic predicate  $K_{A,t}\varphi$ . They are two mathematically distinct routes to iterated self-reference.

## 5 Two kinds of omniscience

**Definition 5.1** (Epistemic logical omniscience). Let  $T$  be a theory whose theorems are enumerated  $\varphi_0, \varphi_1, \varphi_2, \dots$ . An agent  $A$  is epistemically logically omniscient over a domain  $D$  when

$$\forall \varphi \in D (T \vdash \varphi \implies K_A \varphi).$$

This is omniscience in the knowledge coordinate. It does not assert iterated knowledge of that knowledge.

**Definition 5.2** (Reflective omniscience). For the reflection sequence  $\theta_0^A, \theta_1^A, \dots$ , call  $A$  reflectively omniscient at the standard finite level when

$$\forall n \in \mathbb{N}, \quad A \vdash \theta_n^A.$$

Within the existing scale this is equivalent to

$$\text{lev}(A) = \omega.$$

**Remark 5.3** (Qualified use of “omniscience”). Level  $\omega$  is not truth-theoretic omniscience about oneself. It gives closure along the designated provability-reflection tower; it does not provide a truth predicate for every sentence. Thus “reflective omniscience” here means omniscience over the reflection tower, not unrestricted semantic omniscience.

## 6 Hyperfinite epistemic omniscience

**Definition 6.1** (Hyperfinite theorem horizon). Let

$$H \in {}^*\mathbb{N}$$

be unlimited:

$$H > n \quad \text{for every standard } n \in \mathbb{N}.$$

Let  $\{\varphi_n : n \leq H\}$  be a hyperfinite initial segment of a transferred theorem enumeration. Define

$$\text{Omn}_A^K(H) := \bigwedge_{\substack{n \leq H \\ T \vdash \varphi_n}} K_A \varphi_n,$$

understood internally through the corresponding hyperfinite iteration rather than as an external infinitary conjunction. This is hyperfinite epistemic logical omniscience through horizon  $H$ .

**Remark 6.2.** Hyperfiniteness is crucial. The object behaves internally like an extraordinarily long finite construction while  $H$  dominates every standard natural number.

## 7 Hyperfinite reflective omniscience

**Definition 7.1** (Hyperfinite reflection tower). Assume the reflection recursion admits an internal nonstandard extension:

$${}^*\theta_0 = \text{ATP}(\langle A \rangle),$$

$${}^*\theta_{\nu+1} = \text{Pr}(\langle A \rangle, \langle {}^*\theta_\nu \rangle), \quad \nu < H.$$

Define  $\text{Omn}_A^R(H)$  to mean

$$\forall \nu \leq H, \quad {}^*A \vdash {}^*\theta_\nu.$$

This is reflective omniscience through the hyperfinite horizon  $H$ .

**Theorem 7.2** (Restriction theorem). *If  $\text{Omn}_A^R(H)$  holds for unlimited  $H$ , then*

$$\text{lev}(A) = \omega$$

*for the original standard reflection scale, provided standard reflection formulas and their proof relation are preserved under the chosen extension.*

*Proof.* Every standard  $n \in \mathbb{N}$  satisfies  $n \leq H$ . Hence the hyperfinite hypothesis includes every standard reflection formula. Therefore  $A \vdash \theta_n$  for every standard  $n$ , so  $L(A) = \mathbb{N}^+$  and  $\text{lev}(A) = \omega$ .  $\square$

**Corollary 7.3** (Standard upper envelope). *Every standard finite reflection level lies below the unlimited horizon  $H$ . In this precise sense hyperfinite reflective omniscience provides an upper envelope for all standard finite reflectivity.*

## 8 The converse problem

**Remark 8.1** (No automatic reverse transfer). From  $\text{lev}(A) = \omega$  one may conclude  $A \vdash \theta_n$  for every standard finite  $n$ . One may not automatically conclude

$${}^*A \vdash {}^*\theta_H$$

for an unlimited hypernatural  $H$ . The phrase “for every standard  $n$ ” is external when it quantifies specifically over standard integers. Transfer becomes available only after the property has been represented by a suitable internal formula.

**Theorem 8.2** (Conditional reverse-transfer theorem). *Suppose there exists an internal formula  $R(n, A)$  such that for every standard  $n$ ,*

$$R(n, A) \iff A \vdash \theta_n,$$

*and suppose the standard theory proves the internal universal statement*

$$\forall n \in \mathbb{N} R(n, A).$$

*Then transfer yields*

$$\forall \nu \in {}^*\mathbb{N} {}^*R(\nu, {}^*A).$$

*Consequently,*

$${}^*A \vdash {}^*\theta_\nu$$

*for every hypernatural  $\nu$  for which the transferred coding and proof predicates retain the intended interpretation.*

**Remark 8.3.** This is the natural “transfer there and back” route. The forward direction

$$\text{hyperfinite reflective coverage} \implies \text{lev}(A) = \omega$$

is comparatively direct. The reverse direction requires internalization:

$$\text{lev}(A) = \omega + \text{uniform internal reflection theorem} \implies \text{hyperfinite reflective coverage}.$$

Thus  $\omega$ -level reflection by itself is insufficient; uniform internal closure is the bridge.

## 9 Omniscience and the four oracles

**Definition 9.1** (Four-oracle system). Let

$$\mathcal{O} = (O_P, O_R, O_I, O_C)$$

denote four oracle channels associated respectively with proof, refutation, independence or metalogical settlement, and contradiction or integrity. Each oracle has a certified domain  $D_j$  and answer relation  $O_j(\varphi) = a$ . An oracle answer becomes epistemically usable by  $A$  only through the appropriate verifier and accessibility gate.

**Remark 9.2.** An oracle is an architectural resource. Omniscience is a closure property. They are not identical concepts. Four oracles may be powerful without being omniscient, and an omniscient agent need not be architecturally decomposed into four oracles.

**Definition 9.3** (Oracle coverage). The coverage domain of the four-oracle system is

$$D_{\mathcal{O}} := D_P \cup D_R \cup D_I \cup D_C.$$

**Theorem 9.4** (Oracle-cover criterion for epistemic omniscience). *Let  $E$  be an epistemic theorem domain. Suppose:*

- (1) *every  $\varphi \in E$  belongs to at least one oracle domain, so  $E \subseteq D_{\mathcal{O}}$ ;*
- (2) *every applicable oracle returns a correct certificate;*
- (3) *every such certificate is accepted by  $V$ ;*
- (4) *every accepted oracle record enters  $\text{View}_A$ .*

Then

$$T \vdash \varphi, \varphi \in E \implies K_A \varphi.$$

Thus the four oracles jointly induce epistemic logical omniscience over  $E$ .

*Proof.* For every theorem  $\varphi \in E$ , coverage supplies an oracle. Correctness supplies a valid certificate. Verification and accessibility supply a witness record. The definition of  $K_A$  then yields  $K_A \varphi$ .  $\square$

**Corollary 9.5.** *Oracle count is irrelevant by itself. The relevant quantity is certified coverage:*

$$\bigcup_j D_j.$$

*Four narrow oracles need not approach omniscience; four jointly exhaustive oracles can.*

## 10 Oracles and reflective omniscience

**Lemma 10.1** (Epistemic omniscience can dominate finite reflection). *Suppose the epistemic domain  $E$  contains every reflection formula  $\theta_n^A$ , and suppose  $A$  is epistemically omniscient over  $E$ . Then  $K_A \theta_n^A$  for every theorem  $\theta_n^A$  in the tower. If, in addition, there is a sound bridge*

$$K_A \theta_n^A \implies A \vdash \theta_n^A,$$

*then*

$$\text{lev}(A) = \omega.$$

**Remark 10.2.** The bridge is essential. Knowing a formula and proving it are not definitionally identical in the present framework.

**Theorem 10.3** (Hyperfinite epistemic-to-reflective bound). *Suppose  $H$  is unlimited and the hyperfinite theorem horizon contains every standard reflection formula:*

$$\forall n \in \mathbb{N}, \quad \text{index}(\theta_n^A) \leq H.$$

*If  $\text{Omn}_A^K(H)$  holds and verifier-grounded knowledge of reflection formulas is admissible as proof in the reflection system, then*

$$\text{lev}(A) = \omega.$$

*Thus hyperfinite epistemic omniscience provides an upper envelope containing every standard finite reflective requirement.*

## 11 The asymmetry between the two omnisciences

**Theorem 11.1** (Non-equivalence of omniscience coordinates). *In general,*

$$\text{Omn}^K \not\Longleftrightarrow \text{Omn}^R.$$

*More precisely,*

$$\text{Omn}^K + \text{reflection inclusion} + \text{knowledge-to-proof bridge} \implies \text{Omn}^R$$

*on the corresponding reflection domain, while*

$$\text{Omn}^R \not\Rightarrow \text{Omn}^K$$

*over the full theorem domain.*

*Proof.* Reflective omniscience concerns only the special sequence  $\theta_0, \theta_1, \theta_2, \dots$ . Epistemic logical omniscience concerns an entire theorem domain. A machine may therefore possess every finite reflection level while remaining ignorant of arbitrarily many unrelated mathematical theorems. This agrees with the existing Tarski–Goedel ceiling: Level  $\omega$  is closure over a provability-reflection sequence, not unrestricted truth-theoretic omniscience.  $\square$

**Corollary 11.2.** *The two-dimensional awareness geometry extends naturally to a two-dimensional omniscience geometry:*

$$\mathbf{Omn}(A) = (\text{Omn}^K(A), \text{Omn}^R(A)).$$

*One coordinate measures breadth of certified knowledge; the other measures depth of recursive self-reflection.*



## 12 Hyperfinite oracle federation

**Definition 12.1** (Hyperfinite oracle coverage). For unlimited  $H$ , define

$$D_{\mathcal{O}}(H) = \bigcup_{j \in \{P, R, I, C\}} D_j(H).$$

The oracle federation is hyperfinitely exhaustive through  $H$  when

$$\{\varphi_n : n \leq H, T \vdash \varphi_n\} \subseteq D_{\mathcal{O}}(H).$$

**Theorem 12.2** (Hyperfinite four-oracle omniscience). *If the four oracle domains are hyperfinitely exhaustive through  $H$ , all oracle outputs are verifier-certified, and every accepted output is accessible to  $A$ , then*

$$\text{Omn}_A^K(H).$$

*If the hyperfinite horizon also contains the reflection tower and the knowledge-to-proof bridge applies to reflection formulas, then every standard finite reflection level follows and therefore*

$$\text{lev}(A) = \omega.$$

**Remark 12.3.** This produces the hierarchy

$$\begin{aligned} \text{four oracles} &\longrightarrow \text{certified coverage} \longrightarrow \text{hyperfinite epistemic omniscience} \\ &\longrightarrow \text{standard reflective upper bound.} \end{aligned}$$

where every arrow requires stated hypotheses. No arrow should be reversed automatically.

## 13 The Picard bridge

Under the Picard idealization, self-identification together with internal necessitation gives

$$\text{lev}(\text{Picard}) = \omega.$$

This is reflection generated vertically:

$$\theta_0 \rightarrow \theta_1 \rightarrow \theta_2 \rightarrow \cdots$$

The oracle construction is naturally horizontal:

$$O_P, \quad O_R, \quad O_I, \quad O_C,$$

expanding the breadth of propositions for which warranted answers can be obtained.

Accordingly the awareness plane has a natural interpretation:

horizontal coordinate = epistemic breadth

and

vertical coordinate = reflective depth

The Picard mechanism principally moves vertically. The oracle federation principally moves horizontally. A reflective AMLD equipped with a broad oracle federation may move in both directions.

## 14 Final geometric formulation

**Definition 14.1** (Awareness-omniscience plane). Let

$$\mathfrak{A} = \{(a, \lambda) : a \in \mathcal{E}, \lambda \in \omega + 1\},$$

where  $\mathcal{E}$  is the chosen ordered scale of verifier-grounded epistemic awareness and

$$\omega + 1 = \{0, 1, 2, \dots, \omega\}$$

is the reflection-level scale. Associate to each suitable system  $A$  the point

$$\Psi(A) = (aw(A), \text{lev}(A)).$$

**Theorem 14.2** (Independence principle). *Neither coordinate determines the other without additional bridge axioms.*

**Corollary 14.3.** *The awareness state of a formal system is naturally at least two-dimensional.*

**Remark 14.4.** The geometry explains why the two forms of omniscience should not be conflated. Maximal vertical position,

$$\text{lev}(A) = \omega,$$

does not imply maximal horizontal knowledge. Maximal horizontal theorem knowledge implies maximal standard vertical reflection only when reflection formulas lie inside the epistemic domain and the appropriate knowledge-to-proof bridge is available.

**Remark 14.5** (NSA extension). Nonstandard analysis potentially adds a second scale beyond the standard plane. An unlimited hypernatural horizon  $H$  can contain every standard finite epistemic or reflective stage while remaining an internal hyperfinite object. This suggests schematically

$$\boxed{\text{standard awareness plane} \subset \text{hyperfinite awareness geometry}},$$

but establishing equality, transfer, conservation, or reverse extraction requires separate theorems.

**Remark 14.6** (Final distinction). The conceptual distinction can be stated compactly:

$$\boxed{aw(A) \text{ asks what the system knows about itself and its world}}$$

while

$$\boxed{\text{lev}(A) \text{ asks how far it can iterate the fact of its own proving.}}$$

The four oracles enlarge the first coordinate by enlarging certified epistemic reach. Internal necessitation enlarges the second by propagating reflection. Nonstandard analysis supplies a hyperfinite horizon capable, under appropriate internality assumptions, of simultaneously dominating every standard finite stage of either process.

## Source note

The definitions of self-theory, reflection sequence, reflective level, Level  $\omega$ , the Picard idealization, and the Tarski–Goedel ceiling are synthesized from *Depths of Induction and Formalized Self-Awareness, Version 46*. The AMLD federation, verifier-grounded knowledge predicate, structural self-identification route, receipt reflection, and the target “I know that I know that I am an AMLD” are synthesized from *Verifier Certificate 001: The Self Aware Machine*. The two-coordinate awareness geometry, the qualified omniscience terminology, the hyperfinite transfer results, and the oracle-coverage theorems are new conditional extensions built from those source frameworks.